

OBJECT-ORIENTED SOFTWARE ENGINEERING

IMPORTANT QUESTIONS – REGULATION 2025 (R2025) | CS25C10

Topic-wise question bank for examination preparation

Prepared strictly from the uploaded OOSE R2025 important notes, following the topic-wise exam-question style of the uploaded Data Structures Important Questions PDF.

Tip: Prepare definitions and short questions first. For long answers, practice diagrams, comparisons, examples and step-by-step explanations.

UNIT I – SOFTWARE PROCESS AND AGILE DEVELOPMENT

1. Define software engineering. Explain its objectives.
2. Explain software process and its typical activities.
3. Explain Waterfall, Incremental, Prototyping, Spiral and Evolutionary process models.
4. Compare Waterfall model and Agile process.
5. Explain Agile process and its characteristics.
6. Explain the four values of the Agile Manifesto.
7. Explain Extreme Programming (XP) practices and advantages.

UNIT II – REQUIREMENTS ANALYSIS AND SPECIFICATION

1. Define requirements. Differentiate functional and non-functional requirements.
2. Explain Software Requirements Specification (SRS) and its characteristics.
3. Explain formal specification and its importance.
4. Explain Finite State Machine (FSM) with its components and applications.
5. Explain Petri Nets and their use in concurrency, synchronization and resource sharing.
6. Define UML. Explain its purpose in software development.
7. Explain Use Case Diagram and the relationships association, include and extend.
8. Explain Class Diagram and its major relationships.
9. Explain Sequence Diagram and Communication Diagram.
10. Explain Activity Diagram and State Chart Diagram.
11. Explain Data Flow Diagram (DFD) and its components.
12. Explain CASE tools and their applications.

UNIT III – SOFTWARE DESIGN

1. Define software design. Explain the goals of good software design.
2. Explain design metrics: complexity, coupling, cohesion, size and maintainability.
3. Explain modularity and its importance.
4. Define coupling. Explain the different types of coupling.
5. Define cohesion. Explain the different types of cohesion.
6. Compare coupling and cohesion.
7. Compare top-down design and bottom-up design.
8. Explain hybrid design.
9. Differentiate functional-oriented design and object-oriented design.
10. Explain abstraction, encapsulation, inheritance and polymorphism in object-oriented design.

UNIT IV – DESIGN PATTERNS AND ARCHITECTURAL STYLES

1. Define design pattern. Explain its purpose and characteristics.
2. Explain MVC pattern with its Model, View and Controller.
3. Explain Observer pattern with a suitable example.
4. Explain Strategy pattern.
5. Explain Adapter pattern.
6. Explain Command pattern and its applications.
7. Explain Facade pattern.
8. Explain Proxy pattern.
9. Explain Publish-Subscribe pattern.
10. Explain Layered, Client-Server, Tiered and Pipe-and-Filter architectural styles.
11. Explain user interface design principles.
12. Explain the case study approach for software design.

UNIT V – SOFTWARE TESTING AND MAINTENANCE

1. Define software testing. Explain its purpose.
2. Explain Unit Testing.
3. Compare Black-Box Testing and White-Box Testing.
4. Explain equivalence partitioning and boundary value analysis.
5. Explain statement, branch and path-oriented testing.
6. Explain Integration Testing and System Testing.
7. Explain Regression Testing.
8. Explain debugging and its role in software testing.
9. Explain Program Analysis.
10. Explain Symbolic Execution.
11. Explain Model Checking.
12. Define software maintenance. Explain its need.
13. Explain corrective, adaptive, perfective and preventive maintenance.

UNIT VI – PROJECT MANAGEMENT AND DEVOPS

1. Define software project management. Explain its major activities.
2. Explain Software Configuration Management (SCM).
3. Explain configuration items, version control, baselines, change control and status accounting.
4. Explain project scheduling and its important tools.
5. Explain Gantt Chart with its uses.
6. Explain PERT and its three time estimates.
7. Explain Critical Path Method (CPM).
8. Differentiate PERT and CPM.
9. Explain critical path and scheduling slack.
10. Define DevOps. Explain its objectives and characteristics.
11. Explain cloud as a platform for software development and deployment.
12. Explain deployment pipeline.
13. Explain Continuous Integration (CI) and Continuous Delivery/Deployment (CD).
14. Explain the overall DevOps architecture and lifecycle.

EXAM-FOCUSED 16-MARK QUESTIONS

1. Explain software process models and compare Waterfall with Agile. Discuss Agile values and XP practices.
2. Explain requirements analysis and SRS. Discuss functional/non-functional requirements and UML models.
3. Explain Use Case, Class, Sequence/Communication, Activity and State Chart diagrams with suitable examples.
4. Explain coupling and cohesion with their types. Compare top-down, bottom-up and object-oriented design approaches.
5. Explain design patterns: MVC, Observer, Strategy, Adapter, Command, Facade and Proxy.
6. Explain architectural styles and user interface design principles.
7. Compare Black-Box and White-Box Testing. Explain testing levels, techniques and regression testing.
8. Explain debugging, program analysis, symbolic execution and model checking.
9. Explain software maintenance and its four major types.
10. Explain SCM, project scheduling, Gantt Chart, PERT and CPM with suitable examples.
11. Explain DevOps, CI/CD and the deployment pipeline.
12. Explain cloud as a platform and the overall DevOps architecture.

QUICK EXAM REVISION

1. UNIT I: Software process models, Agile values/principles, XP.
2. UNIT II: SRS, functional/non-functional requirements, UML diagrams, FSM, Petri Nets, DFD.
3. UNIT III: Modularity, coupling, cohesion, top-down/bottom-up, object-oriented design.
4. UNIT IV: MVC, Observer, Strategy, Adapter, Command, Facade, Proxy, architectural styles.
5. UNIT V: Black-box/white-box, testing levels, regression, debugging, symbolic execution, model checking, maintenance.
6. UNIT VI: SCM, Gantt, PERT, CPM, critical path, DevOps, CI/CD, deployment pipeline, cloud platform.
7. Long-answer format: Definition → diagram/steps → explanation → example → advantages/applications → conclusion.